

Highlights

Multistage stochastic optimization for fair electricity planning in collective self-consumption communities

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- Multistage stochastic model for fair electricity allocation.
- Fairness ensured via proportional and max–min criteria.
- Photovoltaic and savings allocation policies compared.
- Efficiency–equity trade-offs analyzed under uncertainty.

Multistage stochastic optimization for fair electricity planning in collective self-consumption communities

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Abstract

We study the fair electricity supply planning problem for collective self-consumption communities under uncertainty in photovoltaic generation and electricity demand. We propose a multistage stochastic programming framework that captures the sequential revelation of uncertainty and enables adaptive operational decisions over time. Fairness is embedded directly into the optimization model through two classical criteria, proportional and max-min fairness, applied to two alternative resource definitions: photovoltaic electricity and economic savings. Within this framework, we provide a structural analysis of fairness policies. In particular, we characterize conditions under which proportional and max-min allocations coincide, and derive sensitivity results showing how the efficiency loss induced by fairness evolves with system parameters. These results establish ordering relations between fairness policies and identify regimes in which additional flexibility reduces the cost of fairness. Our computational study, based on real-world photovoltaic data and realistic demand profiles, complements the theoretical analysis. The results show that enforcing fairness induces a non-negligible efficiency loss, whose magnitude depends critically on the resource definition. Policies based on economic savings achieve more equitable outcomes but incur higher efficiency losses, while electricity-based policies lead to lower costs but do not guarantee equitable economic benefits. Finally, we show that system characteristics affect fairness in distinct ways: increasing storage capacity significantly reduces the price of fairness—particularly for savings-based policies—whereas improved demand-generation alignment primarily benefits electricity-based fairness.

Keywords: Energy communities, collective self-consumption, multistage stochastic programming, fair electricity allocation, max-min fairness

1. Introduction

The global transition toward cleaner electricity systems has accelerated the deployment of renewable sources, particularly solar photovoltaic (PV) generation. While these technologies reduce

greenhouse gas emissions and reliance on fossil fuels, their integration poses significant challenges due to their inherent intermittency and dependence on uncertain weather conditions. This variability complicates the real-time balance between supply and demand and, combined with the growing decentralization of generation through rooftop PV and local initiatives, calls for new, more flexible electricity management paradigms beyond traditional centralized models.

In this context, collective self-consumption (CSC) has emerged as a promising approach, enabling groups of co-located prosumers to jointly produce and share renewable electricity. By promoting local use of locally generated electricity, CSC reduces dependence on external supply, limits transmission losses, and supports a more participatory electricity system. However, despite its economic and environmental benefits, CSC also introduces important challenges related to operational management, fair allocation of resources, and sustained community engagement, particularly within regulatory frameworks such as those established in France.

The inherent uncertainties in renewable generation and electricity demand are not static but evolve as dynamic stochastic processes where new information unfolds progressively over time. Weather patterns that affect photovoltaic output change throughout the day as cloud formations develop and dissipate, while electricity demand fluctuates based on occupancy patterns, appliance usage, and behavioral factors that become apparent only as they occur [19]. This dynamic nature of uncertainty means that a collective self-consumption manager cannot simply make all decisions at the beginning of the planning horizon based on forecasts alone. Instead, decision-makers must adapt their strategies dynamically, making operational choices at each point in time based on the most current information available. Thus, a collective self-consumption manager might adjust battery charging schedules as actual solar generation is observed, reallocate electricity among participants as their real consumption patterns emerge or modify grid interaction strategies as updated weather forecasts become available [15].

Beyond handling uncertainty, a central issue in collective self-consumption management is ensuring fairness in the allocation of shared resources and benefits among participants. Fairness is crucial for the long-term sustainability and social acceptance of collective self-consumption communities: if benefits are perceived as unjustly distributed, participants may feel discouraged, potentially leading to community fragmentation or disintegration. Fairness considerations encompass multiple dimensions, including how shared photovoltaic generation is allocated among members, how battery storage is utilized to serve different participants' needs, and how costs and savings are distributed across the community.

The fair electricity supply planning has been investigated in the literature, for example in [8, 10, 13], assuming that inputs such as demand profiles and solar generation are known in advance. However, allocation strategies that appear fair under expected conditions may disproportionately disadvantage some participants when actual conditions deviate from forecasts, and storage usage plans may fail to deliver anticipated benefits if demand peaks unexpectedly or solar output falls short. Thus, the integration of fairness considerations under uncertainty presents theoretical and

computational challenges that distinguish it from fairness in deterministic optimization problems.

Motivated by these challenges, we focus on the fair electricity supply planning problem for collective self-consumption communities under uncertainty through a multistage stochastic programming framework. We investigate two fairness notions from the literature and propose modeling approaches that incorporate fairness considerations into multistage decision making. In particular, we consider alternative definitions of the resource to be allocated, based on photovoltaic electricity and economic savings, which lead to fundamentally different fairness outcomes. In this work, we adopt an ex-ante perspective, which is consistent with sequential planning under uncertainty.

1.1. Literature review

Electricity supply planning in microgrids equipped with photovoltaic panels and battery electricity storage systems (PVB systems) has been widely studied under deterministic assumptions [25, 20, 9, 17, 6, 27]. Early contributions focused on optimal operational scheduling, typically formulated as linear or mixed-integer linear programming models, where photovoltaic production and electricity demand are assumed to be known in advance over the planning horizon. These formulations compute battery charging and discharging schedules, grid interactions, and internal electricity flows to minimize operational costs or improve self-consumption and peak shaving performance.

While a substantial body of literature addresses single-user PVB systems, research on multi-user systems, commonly referred to as collective self-consumption (CSC) communities, remains comparatively limited. Most existing studies consider microgrid communities in which participants independently own distributed energy resources (DERs) and storage systems and are allowed to trade electricity among themselves [24, 18]. In such settings, the primary focus is on pricing mechanisms and market-based coordination strategies regulating internal exchanges. In contrast, building-level CSC communities, where photovoltaic generation and storage are jointly owned and centrally managed and where direct peer-to-peer trading is not permitted, have received less attention. From a technical standpoint, such systems resemble single-user PVB systems because a central entity optimizes shared resources under operational constraints. However, unlike the single-user case, the benefits and costs of shared operation must be allocated among multiple participants, which naturally introduces fairness considerations.

Among deterministic CSC models explicitly addressing fairness, Jorquera-Bravo et al. [13] formulate the electricity supply planning problem with shared photovoltaic generation and storage as a linear program, and analyze classical allocation rules such as proportional fairness and max-min fairness within a centralized optimization framework. Their results show how different allocation rules impact both cost efficiency and the distribution of benefits among participants in a static setting. In addition, Gasca et al. [8] compare centralized and decentralized management schemes, introducing equity-based fairness measures for allocating collective economic benefits under centralized coordination. In a different direction, Gjorgievski et al. [10] propose a virtual net-billing mechanism for collective self-consumption that enables real-time electricity sharing based on in-

dividual contributions. Their approach develops a mathematical framework to evaluate fairness and analyzes the trade-off between fairness and computational scalability across alternative sharing methods.

Despite these contributions, deterministic approaches typically neglect the inherent uncertainty in photovoltaic generation and electricity demand, or reduce fairness to equality-based metrics. In particular, models such as Jorquera-Bravo et al. [13] assume full information over the planning horizon, which limits their ability to capture the dynamic interplay between uncertainty and fairness in real-world operations.

To explicitly account for this uncertainty, stochastic programming approaches have been increasingly adopted in single-user PVB systems. Two-stage models are widely used. For example, Rezaeimozafer et al. [23] model day-ahead grid transactions as first-stage decisions and battery operations as recourse actions, while Zheng et al. [26] separate investment decisions from operational recourse decisions. Multi-stage formulations, such as those proposed by Hafiz et al. [11] and Abedi & Kwon [1], further allow decisions to adapt dynamically as updated information becomes available throughout the planning horizon. In parallel, real-time battery control strategies based on supervised learning have also been explored [14, 16].

When extending stochastic electricity planning to CSC communities, fairness becomes a central design issue. In this context, Erdiñç [7] develops a rolling-horizon MILP framework for real-time energy management in a residential neighborhood with shared PV and ESS resources, incorporating fairness constraints on usage metrics. Uncertainty in photovoltaic production is handled through sequential re-optimization as new information becomes available. While this approach demonstrates that fairness constraints can be integrated with real-time operational decisions at limited cost impact, fairness is ensured through min-max ratio bounds on predefined usage indicators and alternative resource definitions are not analyzed.

Despite these advances, the joint treatment of multistage uncertainty and fairness in collective self-consumption communities remains limited. Existing stochastic CSC studies typically rely on rolling-horizon implementations and focus on a single fairness metric tied to predefined usage indicators. As a result, the combined impact of uncertainty modeling, fairness formulation, and resource allocation rules on both operational efficiency and equity outcomes has not been systematically examined. In particular, it remains unclear how different fairness criteria and allocation principles influence community-level performance when decisions must adapt dynamically to uncertain photovoltaic generation and demand.

Consequently, a comprehensive analysis of how fairness definitions, temporal structure of uncertainty, and resource allocation schemes jointly influence operational efficiency and equity outcomes remains unexplored in the literature. To address these limitations, this work develops a multistage stochastic programming framework for electricity supply planning in collective self-consumption communities and integrates fairness-by-design under different fairness criteria and resource definitions. By comparing different allocation policies, we provide a structured analysis of how fairness

formulations influence operational decisions and cost performance. Through computational experiments based on real-world data, we explicitly quantify the trade-offs between efficiency and equity in stochastic shared-energy systems.

1.2. Contributions

This work makes four main contributions. First, we develop a multistage stochastic programming framework for electricity supply planning in collective self-consumption communities that explicitly captures the sequential revelation of uncertainty. Unlike deterministic or two-stage approaches, the model allows operational and allocation decisions to adapt dynamically over time, providing a realistic representation of energy management under uncertainty. Second, we integrate fairness-by-design into the multistage stochastic framework by embedding fairness constraints directly into the optimization model. We extend two classical fairness criteria, namely proportional and max–min fairness, to a multistage stochastic setting, and consider two alternative resource definitions based on photovoltaic electricity and economic savings. This unified formulation enables a systematic comparison of fairness concepts within a common modeling framework. Third, we provide a structural and sensitivity analysis of fairness policies, showing how the efficiency loss induced by fairness depends on system flexibility and on the economic structure of the model. In particular, we show that savings-based fairness is more strongly coupled with operational decisions, leading to higher efficiency losses but greater sensitivity to flexibility. Fourth, through an extensive computational study based on real-world data, we characterize the joint impact of uncertainty, fairness, and system design. In particular, we characterize the trade-offs between efficiency, equity, and stability, and show how these depend on both the fairness criterion and the resource on which fairness is ensured.

1.3. Paper organization

The remainder of this paper is organized as follows. Section 2 introduces the multistage stochastic formulation of the electricity supply planning problem and describes its scenario-tree representation. In Section 3, we present the fairness criteria considered in this work and describe their integration into the multistage stochastic optimization framework. Section 4 provides a structural and sensitivity analysis of the proposed fairness policies, including results on the efficiency–equity trade-offs. In Section 5, a fix-and-optimize heuristic is developed to address the computational challenges of the model. Section 6 reports the computational experiments based on real-world data and discusses the numerical results. Finally, Section 7 concludes the paper and outlines directions for future research.

2. Stochastic electricity supply planning

We consider residential buildings where households jointly invest in photovoltaic panels and a shared battery. Each household’s demand can be met using PV generation, stored electricity,

or electricity from the main grid. Surplus electricity can be stored or sold to the grid, but not exchanged between users. A central manager allocates PV production over time, and each household is equipped with a smart meter. Simultaneous charging and discharging of the battery, as well as buying and selling electricity in the same period, are not allowed. The system under study is a microgrid consisting of a set $\mathcal{H}$ of smart homes, a photovoltaic installation, a shared battery, and a connection to the external grid. The planning horizon is divided into a set $\mathcal{T}$ of discrete time intervals of duration δ hours, typically $\delta = 1$ hour. The battery is characterized by charging and discharging efficiencies, denoted by γ_c and γ_d , respectively, with $0 < \gamma_c < 1$ and $0 < \gamma_d < 1$, reflecting electricity losses during both processes. It also has an initial state of charge S_0 , a maximum storage capacity $S^{\max}$, and a minimum allowable level $S^{\min}$. Charging and discharging rates are limited by F_c and F_d , respectively. Electricity can be purchased from the grid at a time-dependent price ν_t , discharged from the battery at a unit cost μ , or sold back to the grid at a price β . No operational costs are associated with PV generation, as it does not require fuel or active operation, and maintenance expenses are assumed to be included in the initial investment. The parameters μ and β are constant over time. The selling price β is assumed to be always strictly lower than the purchasing price ν_t for all t .

Electricity supply planning involves inherent uncertainty, mainly from three sources: renewable electricity generation, electricity demand, and market-driven electricity prices [15]. In residential collective self-consumption communities, only the first two sources are relevant, since prices are fixed by pre-existing electricity contracts. To address uncertainty in renewable generation and demand, these factors are modeled as random variables, allowing decisions to be evaluated across multiple possible future scenarios instead of relying on a single forecast. This stochastic approach leads to solutions that are both optimal on average and more robust to changing conditions, motivating the use of stochastic programming techniques [19]. In particular, we adopted a multistage framework, where decisions are made sequentially over time as uncertainty is gradually revealed, enabling adaptive and informed planning.

2.1. Multistage stochastic programming formulation

In practice, uncertainty in photovoltaic generation and demand unfolds gradually, and new information becomes available over time [12]. This enables the community manager to revise allocation and operational decisions dynamically, responding to updated forecasts and observed realizations.

To capture this sequential decision-making process, we extend our framework to a multistage stochastic programming model in which the planning horizon is discretized into stages, each corresponding to a decision point where updated information is incorporated. Formally, we consider a discrete-time stochastic process $\xi_{\sigma=1}^{\Sigma}$ defined on a finite probability space, where Σ denotes the number of decision stages and $\mathcal{S} = 1, \dots, \Sigma$ the set of stages. Each stage $\sigma \in \mathcal{S}$ may comprise one or several consecutive time periods, denoted by $\mathcal{T}^{\sigma}$. The evolution of uncer-

tainty is represented through a discrete scenario tree, and decisions at each stage are required to satisfy non-anticipativity constraints, ensuring that they depend only on the information revealed up to that stage. For each realization of the stochastic process, decisions are defined as $x_\sigma(\xi_\sigma) = (s_t(\xi_\sigma), i_t(\xi_\sigma), g_t(\xi_\sigma), z_t(\xi_\sigma), y_t(\xi_\sigma), p_t(\xi_\sigma), \lambda_t(\xi_\sigma), v_t(\xi_\sigma), w_t(\xi_\sigma))$, $t \in \mathcal{T}^\sigma$, where $s_t(\xi_\sigma)$ denotes the battery state of charge; $i_t(\xi_\sigma)$ the electricity purchased from the main grid by each house; $g_t(\xi_\sigma)$ the electricity sold to the main grid; $z_t(\xi_\sigma)$ and $y_t(\xi_\sigma)$ the amounts of electricity charged to and discharged from the battery, respectively; $p_t(\xi_\sigma)$ the photovoltaic power supplied to each house; and $\lambda_t(\xi_\sigma)$ the share of photovoltaic generation allocated to each house. The binary variables $v_t(\xi_\sigma)$ and $w_t(\xi_\sigma)$ model the activation of battery charging/discharging and electricity purchasing/selling, respectively. Finally, $\xi_\sigma = \{(\mathfrak{C}_t, \{\mathfrak{D}_{j,t}\}_{j \in \mathcal{H}}), t \in \mathcal{T}^\sigma\}$ is a random vector defined on the probability space $(\Omega, \mathcal{F}_\sigma, \mathbb{P})$ with support Ξ_σ . This multistage formulation enables a more realistic representation of collective self-consumption operations, balancing strategic planning with operational adaptability, and provides a richer framework to analyze and compare different fairness formulations in the subsequent sections.

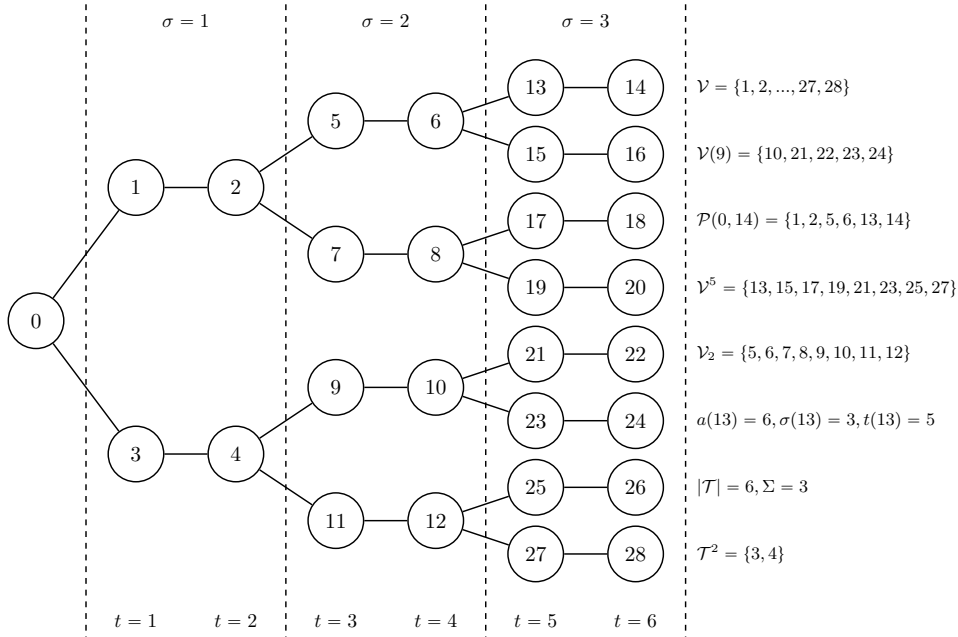


Figure 1: Example of a scenario tree representation of a discrete time stochastic process with $|\mathcal{T}| = 6$ and $\Sigma = 3$.

The information structure is represented by a scenario tree (Figure 1) with node set $\mathcal{V}$. Each node $n \in \mathcal{V}$ is associated with a single time period $t(n)$ and stage $\sigma(n)$, and characterizes the system state given the information available up to $t(n)$. Let $\mathcal{V}^t \subseteq \mathcal{V}$ denote the subset of nodes at time t .

Each node $n \neq 0$ has a unique predecessor $a(n)$ at time $t(n) - 1$, while node 0 denotes the root. For any non-leaf node n , the set $\mathcal{C}(n)$ defines its immediate successors, representing possible realizations of the underlying uncertainty. Let $\mathcal{V}(n)$ denote the subtree rooted at n , and $L(n)$ its

set of leaf nodes.

The probability of node n is denoted by ρ^n . A scenario θ is defined as a path from the root to a leaf node, representing a full realization of the stochastic process over the planning horizon. Finally, $\Theta(n)$ denotes the set of scenarios sharing the node n .

The electricity supply planning problem can be reformulated as a deterministic equivalent model in the form of a MILP as follows:

$$\min_{\{x_n\}_{n \in \mathcal{V}}} \sum_{n \in \mathcal{V}} \rho_n \sum_{j \in \mathcal{H}} c_j(x_n) \quad (1a)$$

s.t.

$$\sum_{j \in \mathcal{H}} \lambda_{j,n} = 1 \quad \forall n \in \mathcal{V} \quad (1b)$$

$$p_{j,n} = \lambda_{j,n} \mathfrak{C}_n \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1c)$$

$$p_{j,n} + y_{j,n} + i_{j,n} - z_{j,n} - g_{j,n} = \mathfrak{D}_{j,n} \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1d)$$

$$s_n = s_{a(n)} + \delta \gamma_c \sum_{j \in \mathcal{H}} z_{j,n} - \frac{\delta}{\gamma_d} \sum_{j \in \mathcal{H}} y_{j,n} \quad \forall n \in \mathcal{V} \quad (1e)$$

$$\sum_{j \in \mathcal{H}} y_{j,n} \leq F_d \quad \forall n \in \mathcal{V} \quad (1f)$$

$$\sum_{j \in \mathcal{H}} z_{j,n} \leq F_c \quad \forall n \in \mathcal{V} \quad (1g)$$

$$S^{\min} \leq s_n \leq S^{\max} \quad \forall n \in \mathcal{V} \quad (1h)$$

$$s_0 = s_n = S_0 \quad \forall n \in \mathcal{V} : t(n) = |\mathcal{T}| \quad (1i)$$

$$z_{j,n} \leq F_c v_{j,n} \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1j)$$

$$y_{j,n} \leq F_d (1 - v_{j,n}) \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1k)$$

$$i_{j,n} \leq M (1 - w_{j,n}) \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1l)$$

$$g_{j,n} \leq M w_{j,n} \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1m)$$

$$v_{j,n}, w_{j,n} \in \{0, 1\} \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1n)$$

$$s_n, i_{j,n}, g_{j,n}, z_{j,n}, y_{j,n}, p_{j,n} \geq 0 \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (1o)$$

Equation (1a) represents the total expected cost of the community, where the individual cost of each house j at each node n is given by Equation (2).

$$c_j(x_n) = \delta(\mu y_{j,n} + \nu_{t(n)} i_{j,n} - \beta g_{j,n}) \quad \forall j \in \mathcal{H}, n \in \mathcal{V} \quad (2)$$

Equations (1b) indicate that at each time period, the proportion of photovoltaic electricity allocated to all houses must be equal to 1. Equations (1c) represent the amount of photovoltaic electricity allocated to each house j at each node $n \in \mathcal{V}$. Constraints (1d) represent the electricity

balance in the community at each node $n \in \mathcal{V}$. Equations (1e) indicate the state of charge of the battery at the end of each node n , which depends on the state of charge of the parent node $a(n)$ at the scenario tree, the amount of electricity charged and the amount discharged in the same node. Constraints (1f) and (1g) represent the limits in the charge and discharge of the battery at each node $n \in \mathcal{V}$. Constraints (1h) set the capacity limits for the state of charge of the battery at each node $n \in \mathcal{V}$. Constraint (1i) imposes that the state of charge of the battery at the end of the time horizon must be equal to the initial state of charge. Constraints (1j) and (1k) indicate that for each house, at each time period, a charge and a discharge of the battery cannot take place at the same time. Similarly, Constraints (1l) and (1m) indicate that a house cannot purchase and sell electricity to the grid in the same node, where M is a large enough parameter.

An important modeling aspect concerns the representation of battery operation. In deterministic settings, Jorquera-Bravo et al. [13] show that binary variables enforcing mutually exclusive charging and discharging decisions can be removed without loss of optimality, leading to a linear programming formulation. In a multistage stochastic setting, such results do not generally extend in a straightforward manner due to the presence of non-anticipativity constraints and the coupling of decisions across scenarios. However, in our formulation, the same structural property holds.

In particular, we show that the binary variables and constraints enforcing mutually exclusive charging/discharging and buying/selling decisions can be omitted without loss of optimality. This leads to a relaxed formulation, denoted by (MS-L), in which all decision variables are continuous. Thus, Proposition 1 establishes dominance properties for any optimal solution to (MS-L). The result follows from the fact that the arguments used in the deterministic case apply independently at each node of the scenario tree, and do not depend on non-anticipativity constraints.

Proposition 1. *Given an instance of the stochastic electricity supply planning problem, where $0 < \gamma_c < 1$, $0 < \gamma_d < 1$, and (ii) $\beta < \nu_{t(n)}$ for each $n \in \mathcal{V}$, any optimal solution $(\{x_n\}_{n \in \mathcal{V}})$ to model (MS-L) satisfies $i_{j,n}g_{j,n} = 0$ and $y_{j,n}z_{j,n} = 0$ for all $j \in \mathcal{H}, n \in \mathcal{V}$.*

Proof. At each node $n \in \mathcal{V}$ of the scenario tree, model (MS-L) has the same structure as the deterministic linear programming model presented in [13] at each time period $t \in \mathcal{T}$. Thus, the result follows by applying the same arguments as in Lemma 1 and Lemma 2 of [13], replacing time periods with scenario tree nodes. $\square$

Proposition 1 implies that the multistage stochastic model (1) admits an equivalent linear programming formulation, as the integrality constraints can be relaxed without affecting optimality.

In the next section, we introduce fairness considerations and show how they can be integrated into this model.

3. Fairness in stochastic electricity supply planning

In stochastic environments, fairness can be evaluated at different moments relative to the realization of uncertainty. A common distinction is between ex-ante and ex-post fairness [2, 3]. Ex-ante fairness assesses allocations before uncertainty is resolved and requires that fairness criteria be satisfied in expectation given the information available at the planning stage. In contrast, ex-post fairness demands that fairness be satisfied in every realized scenario, which constitutes a stronger and typically more restrictive requirement.

In this work, we adopt an ex-ante perspective, which is consistent with multistage stochastic planning where decisions are taken sequentially under uncertainty. This approach enables fairness to be incorporated at the design stage while preserving operational flexibility. Within this framework, we examine two classical fairness concepts, proportional fairness and max–min fairness, and show how they can be embedded into the multistage stochastic electricity supply planning problem. For each fairness criterion, we consider two types of shared resources: Photovoltaic Electricity Allocation (PEA) and Savings Allocation (SA).

3.1. Expected proportional allocation

A classical fairness criteria is the principle of proportionality. According to this view, each participant should receive an amount of the resource that is proportional to their claim [5]. In the following, we extend the concept of proportional fairness to the multistage stochastic electricity supply planning problem.

Given a set of users $j = \{1, \dots, J\}$, let $r_{j,t}(\xi_\sigma)$ denote the amount of resource allocated to user j at time period t when the uncertainty realization up to stage σ is ξ_σ . Let $\mathcal{L}_{j,t}(\xi_\sigma)$ denote the (random) claim of user j at time t , for instance their electricity demand. The cumulative allocation and cumulative claim over the whole planning horizon are therefore random variables.

The Expected Proportional Allocation (EPA) requires that, in expectation, each participant receives the same proportion of their cumulative claim. This corresponds to an ex-ante notion of fairness, since the proportionality condition is evaluated before the realization of uncertainty. Formally, EPA requires that

$$\mathbb{E}_\xi \left[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{j,t}(\xi_\sigma) \right] = \mathbb{E}_\xi \left[\varrho(\xi) \sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} \mathcal{L}_{j,t}(\xi_\sigma) \right] \quad \forall j = 1, \dots, J. \quad (3)$$

Here, the stochastic coefficient $\varrho(\xi)$ represents the proportion of total available resources to total claims under realization ξ . Therefore, EPA ensures that each participant receives, on expectation, a share of the resource proportional to their claim, even though individual realizations may differ across scenarios.

To implement EPA in the stochastic framework, we use the scenario tree representation introduced in Section 2.1. Under this representation, the EPA condition can be written as

$$\sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} r_{j,n} = \sum_{\theta \in \Theta} \rho^\theta \varrho^\theta \sum_{n \in \theta} \mathcal{L}_{j,n} \quad \forall j = 1, \dots, J, \quad (4)$$

where the scenario-dependent coefficient ϱ^θ represents the proportion of total available resources to total claims within scenario θ , and is defined as

$$\varrho^\theta = \frac{\sum_{n \in \theta} \mathcal{R}_n}{\sum_{n \in \theta} \sum_{j=1}^J \mathcal{L}_{j,n}} \quad \forall \theta \in \Theta. \quad (5)$$

From the community manager's perspective, EPA offers a balanced and statistically fair outcome on average, making it attractive when only probabilistic information is available at the planning stage. To incorporate proportional fairness into the electricity supply planning problem, we consider two alternative definitions of the shared resource: PEA and SA.

3.1.1. Proportional photovoltaic electricity allocation

We first consider PEA, where the shared resource is the photovoltaic electricity generated within the collective self-consumption community. Thus, the resource to be allocated at each node $n \in \mathcal{V}$ is $\mathcal{R}_n = \mathfrak{C}_n$, and the claim for each house j at each node $n \in \mathcal{V}$ is $\mathcal{L}_{j,n} = \mathfrak{D}_{j,n}$. Then, the expected proportional (PEA) policy is defined as follows:

$$\sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} p_{j,n} = \sum_{\theta \in \Theta} \rho^\theta \frac{\sum_{n \in \theta} \mathfrak{C}_n}{\sum_{k \in \mathcal{H}, n \in \theta} \mathfrak{D}_{k,n}} \sum_{n \in \theta} \mathfrak{D}_{j,n} \quad \forall j \in \mathcal{H} \quad (6)$$

To implement the PEA policy, constraint (6) is incorporated into the multistage stochastic model (1). This constraint ensures that, in expectation, each house receives a share of the total photovoltaic generation proportional to its aggregate demand over the scenario tree.

3.1.2. Proportional savings allocation

We consider an alternative resource definition: SA, where the shared resource corresponds to the economic savings generated by the collective operation of the community. To quantify these savings, we introduce an all-grid benchmark policy. In this policy each house satisfies its electricity demand exclusively by purchasing electricity from the grid, without participating in the collective self-consumption scheme.

Under this benchmark policy, the cost incurred by house j at node $n \in \mathcal{V}$ is given by $\mathcal{C}_{j,n}^G = \delta \nu_{t(n)} \mathfrak{D}_{j,n}$, where $\nu_{t(n)}$ denotes the electricity purchase price at time period $t(n)$. The total community cost under the all-grid benchmark is therefore $\mathcal{C}_n^G = \sum_{j \in \mathcal{H}} \mathcal{C}_{j,n}^G$. For any feasible solution x_n of the stochastic electricity supply planning model, the total savings generated at node n correspond to the difference between the benchmark cost and the realized operational cost, i.e., $\mathcal{R}_n = \mathcal{C}_n^G - \sum_{j \in \mathcal{H}} c_j(x_n)$, where $c_j(x_n)$ denotes the realized cost of house j . In this setting, the claim of each house is naturally defined as its benchmark cost, i.e., $\mathcal{L}_{j,n} = \mathcal{C}_{j,n}^G$.

Using these definitions, the expected proportional (SA) policy can be written as follows:

$$\sum_{\theta \in \Theta} \rho^\theta \left(\sum_{n \in \theta} (\mathcal{C}_{j,n}^G - c_j(x_n)) \right) = \sum_{\theta \in \Theta} \rho^\theta \left(\frac{\sum_{n \in \theta} \mathcal{C}_n^G - \sum_{k \in \mathcal{H}} c_k(x_n)}{\sum_{n \in \theta} \mathcal{C}_n^G} \sum_{n \in \theta} \mathcal{C}_{j,n}^G \right) \quad \forall j \in \mathcal{H} \quad (7)$$

Constraint (7) is then added to model (1). This constraint ensures that, in expectation, the savings received by each house are proportional to the cost it would incur under the all-grid benchmark policy.

While proportional allocation ensures that resources are distributed according to users' claims, it does not control disparities in the benefits received by different participants. To address this limitation, we also consider a fairness criterion based on the max-min principle, which prioritizes the welfare of the worst-off participant.

3.2. Expected max-min fairness

Another classical fairness criterion is the principle of max-min fairness, which seeks to maximize the minimum benefit received by any participant [5]. Under this principle, the allocation improves the outcome of the worst-off user as much as possible, and no user can be made better off without decreasing the allocation of another user with an equal or smaller benefit.

Thus, the objective is to maximize the smallest expected cumulative allocation across participants over the planning horizon. This is achieved through the following lexicographic optimization model:

$$\text{Lex max}_r \quad \left(\mathbb{E} \left[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[1],t} \right], \mathbb{E} \left[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[2],t} \right], \dots, \mathbb{E} \left[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[J],t} \right] \right) \quad (8a)$$

s.t.

$$0 \leq \sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{j,t}(\xi_\sigma) \leq \sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} \mathcal{L}_{j,t}(\xi_\sigma) \quad \forall j, \xi \in Z \quad (8b)$$

$$\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} \sum_{j=1}^J r_{j,t}(\xi_\sigma) = \sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} \mathcal{R}_t(\xi_\sigma) \quad \forall \xi \in Z \quad (8c)$$

Model (8) computes a lexicographic max-min allocation of the expected cumulative resource received by each participant. The objective maximizes the smallest expected allocation and subsequently improves the allocations of the remaining participants in increasing order. In Equation (8a), $\mathbb{E}[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[1],t}] := \min\{\mathbb{E}[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{1,t}], \dots, \mathbb{E}[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{J,t}]\}$ is the smallest allocation in the allocation vector, $\mathbb{E}[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[2],t}]$ is the second smallest allocation, and so on, so that $\mathbb{E}[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[1],t}] \leq \mathbb{E}[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[2],t}] \leq \dots \leq \mathbb{E}[\sum_{\sigma \in \mathcal{S}} \sum_{t \in \mathcal{T}^\sigma} r_{[J],t}]$. Constraint (8b) ensures that allocations do not exceed participants' cumulative claims under each realization of uncertainty, while Constraint (8c) guarantees that the total available resource is fully distributed. The model is

solved using the direct lexicographic approach proposed in [21], which does not require convexity of the feasible region.

The implementation of the max-min fairness criterion depends on the type of resource being allocated. When allocating photovoltaic electricity, the available resource at each node is exogenous and determined by the photovoltaic generation $\mathfrak{C}_n$. Consequently, the max-min fair benchmark allocation can be computed independently and subsequently ensured within the stochastic electricity supply planning model. In contrast, when the resource corresponds to economic savings, the amount of available resource depends on the operational decisions of the planning model itself. Therefore, the max-min fairness allocation cannot be computed separately, and the lexicographic max-min optimization must be solved directly on the stochastic electricity supply planning model. In the following subsections, we show how this expected max-min fairness formulation can be incorporated into the electricity supply planning problem for the two resource definitions introduced above.

3.2.1. Max-min photovoltaic electricity allocation

As in the previous subsection, we rely on the scenario tree representation of the stochastic process. To obtain a max-min fair allocation of photovoltaic electricity, we first compute the expected aggregate allocation received by each house by solving the following lexicographic max-min optimization model:

$$\text{Lex max}_p \left(\sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} p_{[1],n}, \sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} p_{[2],n}, \dots, \sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} p_{[|\mathcal{H}|],n} \right) \quad (9a)$$

s.t.

$$0 \leq \sum_{n \in \theta} p_{j,n} \leq \sum_{n \in \theta} \mathfrak{D}_{j,n} \quad \forall j \in \mathcal{H}, \theta \in \Theta \quad (9b)$$

$$\sum_{j \in \mathcal{H}} p_{j,n} = \mathfrak{C}_n \quad \forall n \in \mathcal{V} \quad (9c)$$

Model (9) computes a benchmark allocation that maximizes the expected cumulative photovoltaic electricity received by the worst-off house and subsequently improves the allocations of the remaining houses in increasing order. Constraint (9b) ensures that the cumulative allocation to each house under any scenario does not exceed its demand, while Constraint (9c) guarantees that the photovoltaic generation available at each node is fully distributed among the houses.

Let $\bar{p}_{j,n}$ denote the optimal solution of model (9). To ensure this max-min fair allocation within the stochastic electricity supply planning problem, we incorporate the following constraints into model (1) :

$$\sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} p_{j,n} = \sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} \bar{p}_{j,n} \quad \forall j \in \mathcal{H} \quad (10)$$

Constraint (10) ensures that the expected cumulative photovoltaic electricity allocated to each house matches the max-min fair benchmark obtained from model (9), while allowing operational

decisions to adapt to the realization of uncertainty.

3.2.2. Max-min savings allocation

We next consider the allocation of economic savings generated by the collective operation of the community. For each house j , the expected savings are defined as

$$r_j^{SA}(x) = \sum_{n \in \mathcal{V}} \rho_n (\mathcal{C}_{j,n}^G - c_j(x_n)),$$

where $\mathcal{C}_{j,n}^G$ denotes the cost incurred under the all-grid benchmark and $c_j(x_n)$ represents the operational cost obtained from the stochastic electricity supply planning model.

Under this resource definition, the general max-min formulation in Eq. (8) simplifies. In particular, Constraint (8b) reduces to $\sum_{n \in \theta} c_j(x_n) \geq 0 \forall j, \theta$, which ensures that the realized cost of each house remains non-negative in every scenario. Constraint (8c) is satisfied by construction because total savings correspond to the difference between the all-grid benchmark cost and the operational cost obtained from the model.

Using these definitions, the expected max-min fair allocation of savings can be obtained by solving the following lexicographic max-min optimization problem:

$$\text{Lex max}_x \{ (r_{[1]}^{SA}(x), r_{[2]}^{SA}(x), \dots, r_{[|\mathcal{H}|]}^{SA}(x)) \text{ s.t. (1b) -- (1o), } \sum_{n \in \theta} c_j(x_n) \geq 0 \forall j \in \mathcal{H}, \theta \in \Theta \} \quad (11)$$

The lexicographic objective maximizes the expected savings of the worst-off participant and subsequently improves the savings of the remaining participants in increasing order, ensuring an expected max-min fair distribution of savings within the stochastic electricity supply planning framework.

3.3. Structural analysis

An important question is whether the structural properties of the base model, in particular the dominance results established in Proposition 1, are preserved when fairness constraints are introduced. This is not guaranteed in general, as fairness formulations may introduce additional coupling across scenario tree nodes. This leads to the following observation.

Proposition 2. *The dominance properties in Proposition 1 remain valid for fairness formulations based on photovoltaic electricity allocation.*

In these cases, the fairness constraints preserve the local structure of the model, allowing the same arguments as in Proposition 1 to be applied at each node of the scenario tree. However, this is not the case for all fairness definitions. In particular, for proportional and max-min savings, the fairness constraints introduce coupling across nodes, which prevents a direct extension of Proposition 1.

4. Efficiency and structural analysis of fairness policies

In this section, we analyze the efficiency loss induced by fairness constraints through the classical notion of price of fairness. We characterize instances under PEA in which proportional and max-min fairness coincide. Then we study how the resulting efficiency loss depends on system parameters and on the choice of fairness policy.

4.1. Conditions for equivalence between fairness policies

First, we focus on the photovoltaic electricity allocation (PEA) setting. Let $R_j^{PEA}(x) := \sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} p_{j,n}$ denote the expected cumulative PV allocation to household j . We know by Constraints (1b) and (1c), that any feasible allocation satisfies $\sum_{j \in \mathcal{H}} R_j^{PEA}(x) = \bar{\mathfrak{C}}$, where $\bar{\mathfrak{C}} := \sum_{\theta \in \Theta} \rho^\theta \sum_{n \in \theta} \mathfrak{C}_n$ is the total expected PV resource.

For each scenario $\theta \in \Theta$, we define the proportionality factor $\varrho^\theta := (\sum_{n \in \theta} \mathfrak{C}_n) / (\sum_{k \in \mathcal{H}, n \in \theta} \mathfrak{D}_{k,n})$, which depends only on aggregate PV generation and aggregated demand among the houses. Then Equation (6) can be written as $R_j^{PEA}(x) = \sum_{\theta \in \Theta} \rho^\theta \varrho^\theta \sum_{n \in \theta} \mathfrak{D}_{j,n}$ for all $j \in \mathcal{H}$.

Definition 1 (Equal effective claims instance). *An equal effective claims instance is such that all households have equal effective claims under PEA, i.e., $\sum_{\theta \in \Theta} \rho^\theta \varrho^\theta \sum_{n \in \theta} \mathfrak{D}_{j,n} = \sum_{\theta \in \Theta} \rho^\theta \varrho^\theta \sum_{n \in \theta} \mathfrak{D}_{k,n}$ for all $j, k \in \mathcal{H}$.*

In an equal effective claims instance, for example, when all houses have the same cumulative demand at each scenario, i.e. $\sum_{n \in \theta} \mathfrak{D}_{j,n} = \sum_{n \in \theta} \mathfrak{D}_{k,n}$ for all $j, k \in \mathcal{H}, \theta \in \Theta$. Proportional PEA yields identical expected allocations across households, namely $R_j^{PEA}(x) = \bar{\mathfrak{C}}/|\mathcal{H}|$ for all $j \in \mathcal{H}$.

Proposition 3. *Given an equal effective claims instance, the expected allocation vector induced by the lexicographic max-min PEA model (9) coincides with that imposed by proportional PEA in (6). Consequently,*

$$z^{PEA} = z^{MM-PEA}$$

Proof. In an equal effective claims instance, proportional PEA induces the same allocation $R_j(x) = \bar{\mathfrak{C}}/|\mathcal{H}|$ for all j , i.e. proportional PEA induces the equal-allocation vector.

Now, let $\bar{p}$ be lexicographically optimal for (9). At the first lexicographic level, the quantity $\min_j R_j(\bar{p})$ is maximized. Since $\sum_j R_j(\bar{p}) = \bar{\mathfrak{C}}$, any feasible allocation satisfies $\min_j R_j(\bar{p}) \leq \bar{\mathfrak{C}}/|\mathcal{H}|$. Moreover, the equal-allocation vector is feasible and attains this bound, so the first-level optimal value is $\bar{\mathfrak{C}}/|\mathcal{H}|$. Hence $R_j(\bar{p}) \geq \bar{\mathfrak{C}}/|\mathcal{H}|$ for all j . Since these components also satisfy $\sum_j R_j(\bar{p}) = \bar{\mathfrak{C}}$, strict inequality for some j would force strict inequality in the opposite direction for another household, contradicting the previous bound. Therefore $R_j(\bar{p}) = \bar{\mathfrak{C}}/|\mathcal{H}|$ for all j , so both formulations induce the same expected allocation vector and thus $z^{PEA} = z^{MM-PEA}$. $\square$

This equivalence shows that differences between fairness criteria arise only when households exhibit heterogeneous effective claims under PEA. We now examine whether a similar property holds under the savings allocation (SA) setting.

In contrast to PEA, the resource in SA corresponds to economic savings, which depend on the operational decisions and are therefore endogenous. In particular, while the all-grid policy $\mathcal{C}_{j,n}^G$ is fixed, the realized savings $r_j^{SA}(x)$ depend on x , and the total savings can vary across feasible solutions. As a result, equal claims alone are not sufficient to guarantee equivalence between proportional and max-min allocations.

We identify conditions under which SA can be interpreted as a pure redistribution problem, leading to an equivalent result.

Definition 2 (Equal all-grid instance). *An equal all-grid instance corresponds to the case where all households have identical benchmark costs under the all-grid policy, i.e.,*

$$\sum_{n \in \Theta} \mathcal{C}_{j,n}^G = \sum_{n \in \Theta} \mathcal{C}_{k,n}^G \quad \forall j, k \in \mathcal{H}, \quad \forall \theta \in \Theta.$$

In an equal all-grid instance, all households have identical benchmark costs in each scenario, which is analogous to the equal claims instance for PEA. As a result, proportional SA allocates expected savings equally across households. However, this instance is not enough to ensure the equivalence between both fairness policies.

Assumption 1. *The total expected savings $\sum_{j \in \mathcal{H}} r_j^{SA}(x)$ is invariant over the feasible set.*

Under Assumption 1, operational decisions only affect how savings are distributed across households, but not their aggregate level. In this sense, the SA problem becomes analogous to a fixed-resource allocation problem, as in PEA.

Proposition 4. *Given an equal all-grid instance such that Assumption 1 is satisfied. The proportional SA and lexicographic max-min SA induce the same expected savings vector. Consequently,*

$$z^{SA} = z^{MM-SA}.$$

Proof. Let $S_j(x) := r_j^{SA}(x) = \sum_{n \in \mathcal{V}} \rho_n (\mathcal{C}_{j,n}^G - c_j(x_n))$ denote the expected savings of household j , and let $\bar{S} := \sum_{j \in \mathcal{H}} S_j(x)$, which is constant by assumption. Under equal benchmark costs, the right-hand side of (7) is identical for all households, so proportional SA yields $S_j(x) = S_k(x)$ for all $j, k \in \mathcal{H}$. Since total expected savings satisfy $\sum_{j \in \mathcal{H}} S_j(x) = \bar{S}$, it follows that $S_j(x) = \bar{S}/|\mathcal{H}|$ for all j .

Let $\bar{x}$ be lexicographically optimal for (11). At the first level, $\min_j S_j(\bar{x})$ is maximized. Since $\sum_j S_j(\bar{x}) = \bar{S}$, we have $\min_j S_j(\bar{x}) \leq \bar{S}/|\mathcal{H}|$, and this bound is attained by the equal-savings vector. Hence $\min_j S_j(\bar{x}) = \bar{S}/|\mathcal{H}|$, which implies $S_j(\bar{x}) \geq \bar{S}/|\mathcal{H}|$ for all j . Together with $\sum_j S_j(\bar{x}) = \bar{S}$, this yields $S_j(\bar{x}) = \bar{S}/|\mathcal{H}|$ for all j . Thus both formulations induce the same expected savings vector, and $z^{SA} = z^{MM-SA}$. $\square$

4.2. Price of fairness

In this section, we study how the structural differences between fairness policies translate into efficiency losses. In particular, we analyze how the price of fairness responds to variations in system parameters.

First, let us define the price of fairness as follows:

Definition 3 (Price of fairness). *Let $z^{NF} := \min\{c^\top x : x \in X\}$ and $z^F := \min\{c^\top x : x \in X \cap \mathcal{F}_F\}$ denote the optimal costs of the efficient (unconstrained) problem and the fair problem under policy F , respectively. The excess cost of fairness is $\Delta_F := z^F - z^{NF}$, and the corresponding price of fairness is*

$$\text{PoF}_F := \Delta_F / z^{NF}.$$

Let τ denote a scalar right-hand-side (RHS) parameter (e.g., battery capacity $S^{\max}$), and let $z^{NF}(\tau)$ and $z^F(\tau)$ denote the optimal values of the efficient and fair formulations given τ , respectively. Then, the excess cost of fairness given τ is $\Delta_F(\tau) := z^F(\tau) - z^{NF}(\tau)$. Let $\Pi^F(\tau_1)$ and $\Pi^{NF}(\tau_1)$ denote the optimal value of dual variable Π^F and Π^{NF} related to the constraints involving parameter τ , evaluated at $\tau = \tau_1$. This can be interpreted as marginal reductions in the objective when τ increases.

Lemma 1. *For any $\tau_2 \geq \tau_1$,*

$$\Delta_F(\tau_2) - \Delta_F(\tau_1) \leq (\Pi^{NF}(\tau_1) - \Pi^F(\tau_1))(\tau_2 - \tau_1).$$

Proof. Given τ_1 , we first obtain the optimal values of the efficient and fair formulations: $z^{NF}(\tau_1)$ and $z^F(\tau_1)$. Since τ is a RHS parameter, then the dual feasible region does not depend on τ , so the same dual solutions remain feasible at $\tau_2 \geq \tau_1$. Thus, evaluating the dual objective at τ_2 yields

$$z^F(\tau_2) \geq z^F(\tau_1) - \Pi^F(\tau_1)(\tau_2 - \tau_1), \quad z^{NF}(\tau_2) \geq z^{NF}(\tau_1) - \Pi^{NF}(\tau_1)(\tau_2 - \tau_1).$$

Subtracting the second inequality from the first gives

$$\Delta_F(\tau_2) \leq \Delta_F(\tau_1) + (\Pi^{NF}(\tau_1) - \Pi^F(\tau_1))(\tau_2 - \tau_1),$$

which proves the result. $\square$

The previous result characterizes how changes in τ affect the excess cost of fairness through the difference in marginal values. In particular, it yields a simple condition under which additional flexibility reduces the efficiency loss.

Corollary 1. *If $\Pi^F(\tau_1) \geq \Pi^{NF}(\tau_1)$, then*

$$\Delta_F(\tau_2) \leq \Delta_F(\tau_1), \quad \forall \tau_2 \geq \tau_1.$$

This condition admits a direct interpretation: whenever the parameter τ is more valuable under fairness than under the efficient solution, increasing τ reduces the price of fairness. In particular, when $\tau = S^{\max}$, the price of fairness decreases with storage capacity whenever storage is more valuable under fairness.

We now extend this sensitivity analysis to compare fairness policies directly.

Lemma 2. *Let F and G be two fairness policies. For any $\tau_2 \geq \tau_1$,*

$$(\Delta_F(\tau_2) - \Delta_G(\tau_2)) - (\Delta_F(\tau_1) - \Delta_G(\tau_1)) \leq (\Pi^G(\tau_1) - \Pi^F(\tau_1))(\tau_2 - \tau_1).$$

Proof. Applying the sensitivity bound to both F and G and subtracting yields

$$z^F(\tau_2) - z^G(\tau_2) \leq z^F(\tau_1) - z^G(\tau_1) + (\Pi^F(\tau_1) - \Pi^G(\tau_1))(\tau_2 - \tau_1).$$

Rewriting in terms of excess costs gives

$$(\Delta_F(\tau_2) - \Delta_G(\tau_2)) \leq (\Delta_F(\tau_1) - \Delta_G(\tau_1)) + (\Pi^F(\tau_1) - \Pi^G(\tau_1))(\tau_2 - \tau_1),$$

which implies the result. $\square$

The previous result characterizes how the relative performance of fairness policies evolves with system parameters. In particular, it provides a condition under which one policy becomes relatively more efficient than another as flexibility increases. A direct implication is that, whenever the parameter is more valuable under policy F than under G , the relative excess cost of F decreases faster. As a consequence, policy F becomes relatively more efficient as τ increases.

Corollary 2. *If $\Pi^F(\tau_1) \geq \Pi^G(\tau_1)$, then*

$$\Delta_F(\tau_2) - \Delta_G(\tau_2) \leq \Delta_F(\tau_1) - \Delta_G(\tau_1), \quad \forall \tau_2 \geq \tau_1.$$

Since the price of fairness is a normalized version of the excess cost, these results also provide insight into the behavior of $\text{PoF}_F(\tau)$. In particular, when $\tau = S^{\max}$, policies that rely more heavily on operational flexibility may experience a faster reduction in their price of fairness as storage capacity increases. These results highlight that the price of fairness is driven by how fairness constraints interact with the economic structure of the model, which differs across resource definitions.

This distinction has two main implications. First, the sensitivity results apply directly to PEA and extend to SA only through the linear relaxation, where dual values provide approximate marginal information. Second, because savings-based fairness operates on cost-relevant quantities, it typically exhibits stronger dependence on system flexibility. In particular, in typical operating

regimes, one expects

$$\Pi_{\tau}^{SA} \gtrsim \Pi_{\tau}^{PEA},$$

so that savings-based fairness may induce higher efficiency losses at low levels of flexibility, but its excess cost decreases more rapidly as flexibility increases.

5. Fix-and-optimize heuristic

Although the stochastic model admits a linear programming formulation, solving the multistage problem remains computationally challenging due to the exponential growth of the scenario tree. In particular, battery decisions in model (1) induce intertemporal and cross-scenario coupling, which significantly enlarges the effective decision space and leads to long solution times, especially for savings-based and lexicographic formulations. Moreover, the introduction of fairness constraints may further alter the structural properties of the model.

To address these challenges, we propose a fix-and-optimize heuristic that exploits the structural properties of the base model MS-L. The key idea is to decompose the decision process into two stages. First, a subset of decisions is optimized without fairness constraints, exploiting the linear structure of the base model. Then, these decisions are fixed, and the remaining variables are optimized to satisfy the fairness constraints.

Concretely, the method relies on a baseline-driven variable fixing strategy. Let $(\bar{y}_{j,n}, \bar{z}_{j,n}) \forall j \in \mathcal{H}, n \in \mathcal{V}$ be the battery discharge and charge decisions obtained from solving model MS-L. These decisions define an efficient baseline operational policy. The proposed method fixes these variables and re-optimizes the remaining decision variables under the fairness constraints of interest.

This strategy is motivated by two considerations. First, battery variables are a primary source of computational complexity, as they couple decisions across time, households, and scenarios. Second, the baseline solution provides an economically meaningful reference policy. Fixing battery operation preserves this structure while substantially reducing the dimensionality of the problem.

From a methodological perspective, the approach can be interpreted as a structured variable-fixing strategy, in which a subset of decisions is fixed according to a baseline policy and the remaining variables are re-optimized. By fixing the battery decisions, the resulting problems remain linear and can be solved to optimality within the reduced feasible set. As a consequence, the method preserves tractability across all fairness formulations, including those for which the dominance properties in Proposition 1 do not hold.

For proportional fairness policies and max-min fairness PEA, the re-optimization step consists of solving model ML-S augmented with the expected proportional allocation constraints, for proportional PEA (6), for proportional SA (7) or for max-min fairness PEA (10). While fixing battery variables to the baseline values. This leads to a single linear program that enforces fairness exactly within the reduced feasible set.

A similar variable-fixing principle extends to max-min SA. After solving model MS-L, the battery decisions are fixed to the baseline values in model (11), by solving this new model we obtain the value of the remaining variables. The resulting model is a lexicographic max-min model with only continuous variable. In contrast to the other fairness policies, max-min SA cannot be enforced through a single LP and requires to be solved using the direct lexicographic approach proposed in [22].

6. Numerical experiments

In this section we present numerical experiments conducted to illustrate the impact of introducing uncertain parameters into the electricity supply planning problem, and into the fair electricity supply planning problem.

6.1. Instance generation

We randomly generate instances for the stochastic electricity supply planning problem. We created instances with one-day time horizon, and time periods of one hour. We consider three different scenario tree structures with $(\Sigma, B, b) = \{(6, 2, 4); (8, 2, 3); (6, 4, 4); (6, 6, 4)\}$, where Σ denotes the number of stages, B denotes the number of equiprobable realizations per stage and, b the number of time periods per stage. These structures induce 32, 128, 1024, 7776 scenarios, respectively. We then create three consumptions profiles to generate the demand of each house. Houses of type 1 only consumes electricity between 00:00 and 07:59. Houses of type 2 only consumes electricity between 08:00 and 15:59. Finally, houses of type 3 only consumes electricity between 16:00 and 23:59. At each time period, the power demand of houses consuming electricity distribute uniform between 4 kW and 9.2 kW.

To simulate the photovoltaic electricity generation we assume an autocorrelated time serie, thus, we use an Autoregressive Moving Average (ARMA) model, with a correlation of one time period for the autoregressive (AR) part and one time period for the moving average (MA) part. At each node, the photovoltaic production is computed as:

$$\mathfrak{C}_n = \max\{C_{t(n)}^{PV}(1 + \tilde{\epsilon}_n), 0\} \quad \forall n \in \mathcal{V}, \quad (12)$$

where $C_{t(n)}^{PV}$ is the deterministic prediction for the photovoltaic production at time period $t(n)$, and $\tilde{\epsilon}_n$ is the realization of the stochastic prediction error factor in node n .

The deterministic prediction for the photovoltaic production is obtained from historical data collected by E4C's DrahiX demonstrator [4]. While, for the stochastic prediction error factor, we assume, that $\tilde{\epsilon}$ follows an ARMA process of order (1,1) with parameters 0.8 and 0.2, as suggested by Ji & Chee [12]. Thus, the realization of the prediction error factors was generated using the

following model:

$$\begin{cases} \tilde{\epsilon}_0 \sim \mathcal{N}(0, 1) & (13a) \\ \epsilon_n \sim \mathcal{N}(0, \sigma) & \forall n \in \mathcal{V} \cap \{0\} & (13b) \\ \tilde{\epsilon}_n = 0.8\tilde{\epsilon}_{a(n)} + 0.2\epsilon_{a(n)} + \epsilon_n & (13c) \end{cases}$$

All other parameters are considered deterministic, and we used the same values as the one described in [13].

6.2. Experimental setup

This section describes the experimental setting used to evaluate the proposed multistage stochastic optimization framework and the different fairness policies. We consider the following optimization models:

- NF: baseline model defined by (1a)–(1o), without fairness constraints, minimizing expected operational cost.
- PEA: model (1a)–(1o) with proportional fairness constraints on photovoltaic allocation, as defined in (6).
- SA: model (1a)–(1o) with proportional fairness constraints on economic savings, as defined in (7).
- MM PEA: model (1a)–(1o) with max–min fairness on photovoltaic allocation, implemented through constraints (10) derived from (9).
- MM SA: model (11), which integrates max–min fairness directly on economic savings within the stochastic optimization problem.
- DET: deterministic counterpart of (1a)–(1o), where uncertain parameters are replaced by their expected values.

All stochastic models are formulated as multistage stochastic programs based on the scenario tree representation described in Section 2.1, ensuring non-anticipativity of decisions. The deterministic model serves as a benchmark to assess the value of explicitly modeling uncertainty.

To evaluate the different components of the proposed framework, we consider scenario tree structures of increasing size, depending on the objective of each experiment. First, to assess the value of the multistage stochastic formulation, we analyze three scenario tree structures with 32, 128, and 1024 scenarios. These instances allow us to quantify the impact of uncertainty while keeping the exact models computationally tractable. Second, to analyze the impact of fairness policies, we focus on a representative scenario tree with 1024 scenarios. This choice provides a

balance between model fidelity and computational tractability, enabling a detailed comparison of fairness outcomes across different criteria. Finally, to evaluate the performance and scalability of the fix-and-optimize heuristic, we consider a larger scenario tree with 7776 scenarios. This setting highlights the ability of the proposed method to handle large-scale instances where solving the exact multistage models becomes computationally challenging. This experimental design allows us to separately evaluate the modeling benefits, fairness implications, and computational performance of the proposed approach.

All models were implemented in Julia 1.8.1 and solved using IBM ILOG CPLEX 22.1. Experiments were conducted on an Intel Xeon CPU E5-2643 (3.50GHz) with 12 threads and 256GB of RAM. Each configuration was evaluated across multiple scenario tree structures and stochastic realizations to ensure robustness of the results.

6.3. Value of the multistage stochastic formulation

This section evaluates the benefit of the multistage stochastic formulation compared to deterministic planning approaches. We compare the deterministic model (DET), the expected value solution (EEV), the multistage stochastic solution (RP), and the perfect information solution (WS).

The value of the stochastic solution corresponds to the percentage of cost decrease obtained by using a stochastic approach. It is computed as $VSS = \frac{EVV - RP}{RP}$. To compute EEV we solve the deterministic model, considering for each house j at each time step t the demand as the expected value of $\mathcal{D}_{j,t}$ and the production of photovoltaic electricity as the deterministic prediction C_t^{PV} . We record the value of the first-stage variables $\bar{\lambda}$ in this solution. Then, we fix $\lambda_{j,n} = \bar{\lambda}_{j,t(n)}$ for all houses j and nodes n . Thus, all nodes representing the same time period t are fixed to the same value of $\bar{\lambda}_t$.

Table 1: Value of the Stochastic Solution (VSS, %) across scenario tree sizes ($|\Theta|$), number of households ($|J|$), and variability levels (σ). Statistics report average, minimum, maximum, and standard deviation over all instances.

		$\sigma = 0.2$				$\sigma = 0.6$			
$ \Theta $	$ J $	Mean (%)	Min (%)	Max (%)	Std	Mean (%)	Min (%)	Max (%)	Std
32	5	4.38	0.00	19.99	4.82	10.30	0.01	44.78	12.12
32	10	1.88	0.12	5.06	1.53	9.95	0.21	29.86	9.69
128	5	3.15	0.16	11.11	3.07	6.94	0.21	24.69	7.21
128	10	2.16	0.02	6.65	1.70	10.49	0.06	29.28	7.94
1024	5	2.97	0.39	6.76	1.69	12.29	0.84	32.56	10.86
1024	10	2.15	0.03	4.82	1.40	7.79	1.03	20.86	5.59

Table 1 reports the Value of the Stochastic Solution (VSS) across different scenario tree sizes, community sizes, and uncertainty levels. The results quantify the impact of uncertainty and system aggregation on the performance of stochastic planning.

First, the VSS increases significantly with the level of uncertainty. When the standard deviation of the photovoltaic forecast error increases from $\sigma = 0.2$ to $\sigma = 0.6$, the average VSS rises from

approximately 2%–4% to 7%–12%, with peak values exceeding 40%. This confirms that the benefit of stochastic optimization is strongly driven by the variability of renewable generation. Under low uncertainty, deterministic solutions based on expected values remain competitive, as forecast deviations can be partially absorbed through operational flexibility (e.g., battery usage and grid exchanges). However, as uncertainty increases, deterministic decisions become systematically misaligned with realized conditions, leading to significant efficiency losses. The multistage stochastic formulation mitigates this effect by allowing decisions to adapt dynamically to the progressive revelation of uncertainty.

Second, the VSS is consistently higher in smaller communities ($|J| = 5$) than in larger ones ($|J| = 10$), particularly under high uncertainty. This effect is observed across all scenario tree configurations. The result can be attributed to a risk-pooling mechanism: larger communities benefit from demand aggregation, which reduces variability in net load and partially hedges against fluctuations in photovoltaic generation. In contrast, smaller communities exhibit greater exposure to supply–demand mismatches, thereby increasing the marginal value of adaptive stochastic decisions.

Third, the dispersion of VSS values across instances, as reflected by the standard deviations and maximum values reported in Table 1, indicates a strong dependence on instance characteristics. In particular, high VSS values are associated with instances combining large forecast errors and weak temporal alignment between demand and generation, where deterministic policies perform poorly.

Finally, the Value of Perfect Information (EVPI) is consistently close to zero across all tested configurations and is therefore not reported. This suggests that the multistage stochastic formulation captures most of the value of uncertainty through recourse decisions.

6.4. Impact of fairness policies

In this section, we analyze the impact of incorporating fairness constraints into the stochastic electricity supply planning problem. While the previous section focused on the value of modeling uncertainty, here we examine how different fairness policies influence both operational efficiency and the distribution of benefits among households. To this end, we evaluate four fairness policies and compare them against the baseline model without fairness constraints. We structure the analysis along three complementary dimensions: (i) the efficiency loss induced by fairness, (ii) the consistency of allocations at the household level, and (iii) the resulting distribution of benefits across participants.

6.4.1. Efficiency of fairness policies

We evaluate the efficiency loss induced by fairness constraints using the price of fairness (see Def. 3). Table 2 reports descriptive statistics of PoF across fairness policies and uncertainty levels, while Figure 2 illustrates its distribution across instances.

All fairness policies induce a strictly positive PoF, confirming that enforcing equity entails an efficiency loss. However, the magnitude of this loss varies substantially across policies and depends

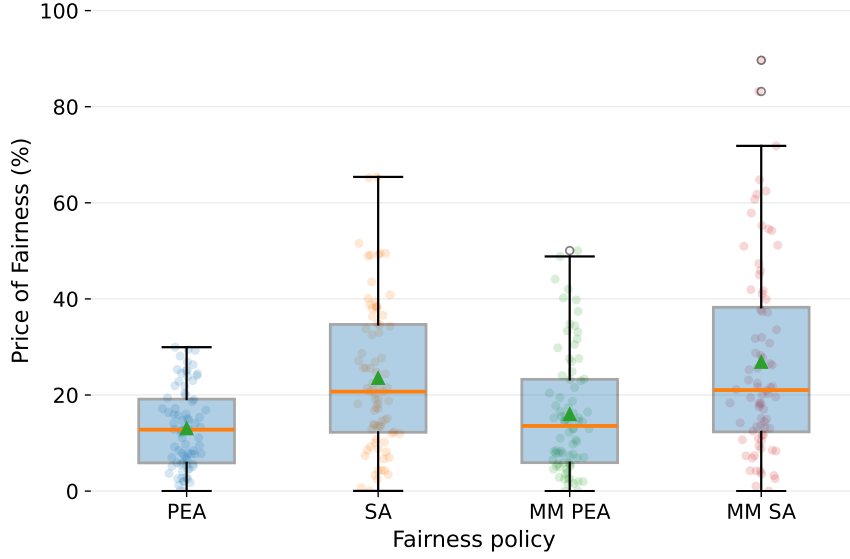


Figure 2: Distribution of the price of fairness (PoF) across instances for each fairness policy and uncertainty level. The figure illustrates how uncertainty affects the efficiency–equity trade-off.

Table 2: Price of fairness (PoF, %) across fairness policies and uncertainty levels (σ). Reported statistics include mean, minimum, maximum, and standard deviation over all instances.

	$\sigma = 0.2$				$\sigma = 0.6$			
Fairness	Mean	Min	Max	Std	Mean	Min	Max	Std
PEA	11.90	0.01	29.45	8.10	14.16	1.79	29.95	8.36
SA	20.26	0.04	49.19	12.60	26.79	4.22	65.39	16.87
MM PEA	13.35	0.00	42.01	10.55	18.63	1.59	50.05	13.81
MM SA	21.37	0.00	57.86	14.17	32.37	4.14	89.66	23.49

on both the fairness criterion and the level of uncertainty.

First, policies based on savings allocation (SA and MM SA) systematically exhibit higher PoF values than their electricity-based counterparts (PEA and MM PEA). As shown in Table 2, the average PoF is significantly larger for savings-based policies, with a wider dispersion across instances, as also reflected in Figure 2. These results suggest that the efficiency loss depends on the level at which fairness is ensured. Savings-based policies tend to induce higher efficiency losses, possibly because fairness is imposed directly on economic outcomes that are closely aligned with the objective function. In contrast, electricity-based policies operate on the allocation of a physical resource, which allows the system to partially recover efficiency through operational adjustments.

Second, max–min fairness policies (MM PEA and MM SA) lead to higher efficiency losses than proportional fairness policies (PEA and SA). This effect is particularly pronounced for MM SA, which consistently exhibits the highest PoF values and the largest variability. This behavior reflects the conservative nature of max–min criteria, which prioritize improving the outcome of the worst-off participant, even when this requires significant deviations from cost-efficient allocations.

Third, the price of fairness increases with the level of uncertainty. When σ increases from 0.2 to 0.6, the average PoF rises across all policies, and the corresponding distributions shift toward higher values. This indicates that uncertainty amplifies the efficiency–equity trade-off: higher uncertainty increases the value of operational flexibility, while fairness constraints limit the system’s ability to adapt to realized conditions.

Finally, the dispersion observed in PoF values, particularly for savings-based and max–min policies, indicates that the cost of fairness is highly instance-dependent. As illustrated in Figure 2, these policies exhibit heavier tails, with some instances incurring very high efficiency losses. These extreme cases typically arise when demand profiles and photovoltaic generation are poorly aligned, forcing the system to rely more heavily on costly grid interactions to maintain fairness.

6.4.2. Efficiency sensitivity to storage and demand

The results in 3 and 4 illustrate how system flexibility and demand–generation alignment affect the efficiency loss induced by fairness constraints.

Table 3: Descriptive statistics of price of fairness (%) under battery scaling (BatteryScale = 2.0)

Fairness	$\sigma = 0.2$				$\sigma = 0.6$			
	Mean	Min	Max	Std	Mean	Min	Max	Std
PEA	7.13	0.00	23.81	6.84	10.61	0.08	29.56	7.73
SA	10.51	0.00	40.44	10.52	13.59	0.02	53.59	12.47
MM PEA	8.25	0.00	33.10	9.04	15.37	0.11	61.56	14.08
MM SA	11.69	0.00	48.77	12.66	19.22	0.04	83.24	19.26

Table 4: Descriptive statistics of price of fairness (%) under high demand–PV alignment (DemandProfile = align_high)

Fairness	$\sigma = 0.2$				$\sigma = 0.6$			
	Mean	Min	Max	Std	Mean	Min	Max	Std
PEA	6.52	0.00	24.85	7.16	6.39	0.00	24.76	5.97
SA	15.99	0.00	59.63	15.71	18.58	0.00	71.53	15.43
MM PEA	6.67	0.00	27.41	7.48	7.21	0.00	34.91	7.81
MM SA	16.17	0.00	61.16	16.20	20.02	0.00	94.00	18.34

Increasing storage capacity (Table 3) leads to a systematic reduction in the price of fairness across all policies. This effect is particularly pronounced for savings-based formulations, whose mean PoF decreases significantly relative to the base case. For instance, SA and MM SA experience substantial reductions in both mean and dispersion, especially under high variability. This pattern is consistent with the sensitivity analysis: since savings-based fairness relies more heavily on operational flexibility, additional storage disproportionately alleviates the cost of enforcing fairness.

In contrast, allocation-based policies also benefit from increased storage, but to a lesser extent. The reduction in PoF for PEA and MM PEA is more moderate, indicating that their efficiency loss

is less sensitive to storage expansion. This suggests that the marginal value of flexibility is higher under savings-based fairness, in line with the theoretical prediction that $\Pi_{\tau}^{SA} \gtrsim \Pi_{\tau}^{PEA}$ in typical regimes.

A different pattern emerges when modifying the demand profile while keeping system capacity fixed (Table 4). Improved alignment between demand and PV generation reduces the price of fairness for allocation-based policies, which exhibit both lower mean values and reduced dispersion relative to the base case. In contrast, savings-based policies remain significantly more costly, with only modest improvements. This indicates that better alignment primarily benefits allocation-based fairness, while savings-based fairness continues to require substantial redistribution of cost-relevant decisions.

Across all scenarios, max–min formulations tend to increase the price of fairness relative to their proportional counterparts, particularly under higher uncertainty. This effect is again more pronounced for savings-based policies, where the combination of cost-based fairness and worst-case prioritization amplifies both efficiency losses and variability.

6.4.3. Equity analysis

To complement the analysis of efficiency and household-level stability, we now assess the distributional impact of fairness policies across households. While previous sections examined aggregate performance and scenario-level variability, this analysis focuses on the structural distribution of benefits within the community. To this end, we evaluate standard inequality metrics computed over the expected savings of each household, including Jain’s index, the Gini coefficient, and the max–min ratio.

Table 5: Equity metrics based on expected photovoltaic electricity allocation across fairness policies and uncertainty levels (σ). Reported values correspond to average Jain index, Gini coefficient, and max-to-min ratio over all instances.

Fairness	$\sigma = 0.2$			$\sigma = 0.6$		
	Jain	Gini	Max/Min	Jain	Gini	Max/Min
NF	0.53	0.48	47.22	0.52	0.48	65.35
PEA	0.98	0.04	1.28	0.96	0.07	1.53
SA	0.66	0.35	7.74	0.67	0.35	9.21
MM PEA	1.00	0.00	1.01	1.00	0.00	1.01
MM SA	0.69	0.34	6.69	0.65	0.38	7.98

Tables 5 and 6 report the average values of the inequality metrics based on photovoltaic allocation and economic savings, respectively. The results show that equity outcomes depend strongly on both the fairness criterion and the resource definition.

A first key observation is the distinction between electricity-based and savings-based fairness. Policies based on photovoltaic allocation (PEA and MM PEA) achieve near-perfect equity in terms of electricity distribution, with Jain indices close to 1 and Gini coefficients close to 0. However, this does not translate into equally equitable economic outcomes. In terms of savings, these policies still

Table 6: Equity metrics based on expected savings across fairness policies and uncertainty levels (σ). Reported values correspond to average Jain index, Gini coefficient, and max-to-min ratio over all instances.

Fairness	$\sigma = 0.2$			$\sigma = 0.6$		
	Jain	Gini	Max/Min	Jain	Gini	Max/Min
NF	0.54	0.46	28.38	0.53	0.48	30.12
PEA	0.71	0.31	6.38	0.67	0.35	11.75
SA	0.93	0.08	1.83	0.85	0.16	3.08
MM PEA	0.74	0.29	5.15	0.73	0.31	6.42
MM SA	1.00	0.00	1.00	1.00	0.00	1.00

exhibit significant inequality, indicating that equalizing physical resource allocation is not sufficient to ensure fairness in economic benefits. In contrast, savings-based policies (SA and MM SA) achieve substantially better equity in economic outcomes. SA significantly reduces inequality, while MM SA leads to near-perfect equality in savings distribution. This highlights that enforcing fairness directly on economic outcomes is more effective in controlling disparities in benefits.

Regarding the fairness criterion, max-min policies induce stronger redistribution effects than proportional ones. While proportional policies preserve some heterogeneity, max-min policies push the system toward more uniform allocations. This effect is particularly pronounced for MM SA, which equalizes savings across households at the cost of higher efficiency losses.

Finally, increasing uncertainty slightly deteriorates equity outcomes across most policies. When σ increases, Jain indices decrease and Gini coefficients increase, indicating that uncertainty introduces additional dispersion in benefits, even when fairness is ensured in expectation.

6.5. Performance of the fix-and-optimize heuristic

The computational results consistently show that the proposed heuristic achieves a highly favorable trade-off between solution quality and computational effort across all fairness measures.

From a solution quality perspective, Table 7 reports that the heuristic remains remarkably close to optimality. The average optimality gap is below 1.1% in all cases and below 0.5% for proportional and max-min PEA policies. Moreover, for PEA-based formulations, more than 60% of instances exhibit an optimality gap below 0.5%, indicating that the heuristic frequently recovers near-optimal solutions.

Figures 3a and 3b show the performance profiles for PEA and MM PEA, respectively. In both cases, the heuristic consistently dominates the exact method, requiring less computational time for nearly all fractions of solved instances. The improvement is moderate but stable, reflecting the relatively well-conditioned structure of these formulations.

In contrast, Figures 4a and 4b highlight a markedly different behavior for savings-based measures. The exact formulation exhibits a heavy computational tail, with solution times increasing sharply for the most difficult instances. The heuristic effectively mitigates this behavior, significantly reducing the time required to solve a large fraction of instances. This effect is particularly

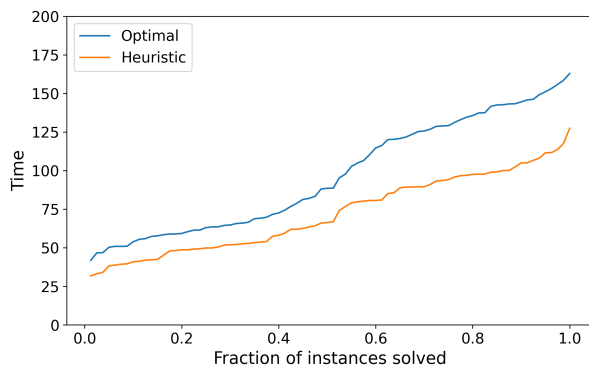
Table 7: Performance comparison of the fix-and-optimize heuristic across fairness criteria. Gap statistics (in %) are computed relative to the optimal solution. Reported times correspond to average solution times for the heuristic and the exact formulation.

Fairness	Gap				Time	
	Mean (%)	Min (%)	Max(%)	Std (%)	Mean Heuristic (s)	Mean Optimal (s)
PEA	0.45	0.00	1.10	0.29	75.2	97.0
SA	1.11	0.00	5.50	1.05	181.5	1035.1
MM PEA	0.50	0.00	1.50	0.35	104.6	149.3
MM SA	1.06	0.00	4.20	0.86	150.4	275.0

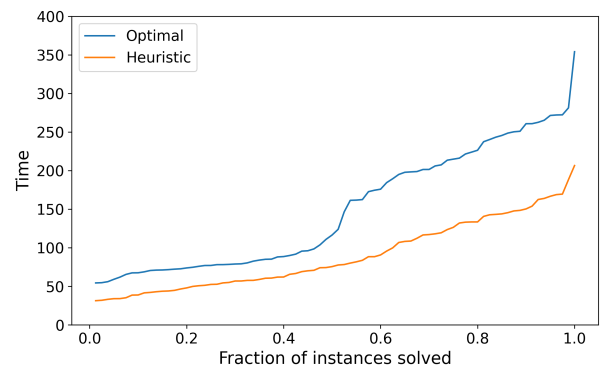
evident in the upper quantiles of the performance profiles. For proportional SA, the time required by the exact method in the highest quantiles is between 4 and 6 times larger than that of the heuristic, whereas for Max–Min SA the gap ranges between approximately 2 and 4 times. These differences are consistent with the summary statistics reported in Table 7, where average solution times are substantially lower for the heuristic, especially in the SA case.

Finally, max–min variants follow similar trends. As shown in Figures 3b and 4b, MM PEA behaves similarly to PEA, with low variability and consistent dominance of the heuristic, while MM SA exhibits improved stability compared to SA, although still benefiting significantly from the proposed approach. These results indicate that the proposed heuristic provides near-optimal solutions at a fraction of the computational cost, with particularly significant gains in the most computationally challenging settings.

From a practical perspective, this makes the approach well suited for applications requiring repeated or time-constrained computations, such as operational planning or scenario analysis. In these contexts, the heuristic offers a reliable surrogate for the exact models, preserving high solution quality while ensuring computational tractability.



(a) Proportional PEA



(b) Max–min PEA

Figure 3: Performance profiles for photovoltaic electricity allocation under proportional and max–min fairness.

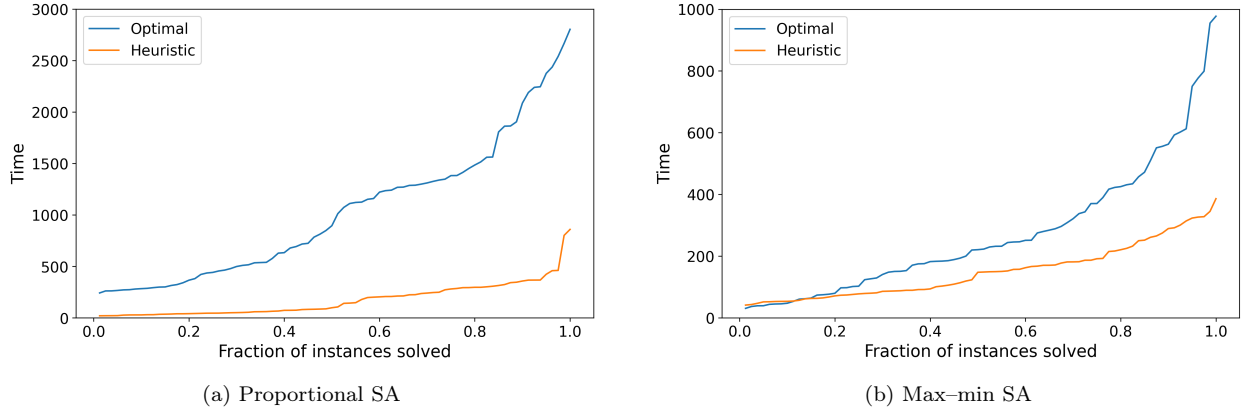


Figure 4: Performance profiles for savings allocation under proportional and max-min fairness.

7. Conclusion

In this work, we studied the stochastic fair electricity supply planning problem for collective self-consumption communities under uncertainty in photovoltaic generation and electricity demand. We proposed a multistage stochastic programming framework that integrates fairness directly into the optimization model under different fairness criteria and resource definitions. The framework was evaluated on instances derived from real-world photovoltaic production data and realistic demand profiles.

The results highlight the importance of explicitly modeling uncertainty. The multistage stochastic formulation provides significant improvements over deterministic approaches, particularly under high variability, while capturing most of the value of uncertainty through adaptive recourse decisions. Our analysis reveals a clear efficiency–equity trade-off. All fairness policies induce a positive cost increase, whose magnitude depends not only on the fairness criterion but also on the resource on which fairness is ensured. In particular, policies based on economic savings lead to higher efficiency losses than those based on electricity allocation.

More broadly, the results show that the effectiveness of fairness policies depends critically on the choice of resource. Enforcing fairness on photovoltaic electricity leads to balanced physical allocations but does not necessarily translate into equitable economic outcomes. In contrast, policies based on economic savings more effectively reduce disparities in benefits, at the cost of reduced efficiency. This highlights that the definition of the resource is a key modeling choice that fundamentally shapes the efficiency–equity trade-off.

These findings open several directions for future research. In particular, designing fairness policies that explicitly control or reduce regret at the household level is a promising direction, although such approaches are likely to introduce additional efficiency losses. More generally, exploring formulations that bridge the gap between ex-ante and ex-post fairness in multistage settings is of particular interest. While ex-post fairness provides stronger guarantees, it is inherently more

restrictive and may lead to significant efficiency deterioration. Developing intermediate or adaptive fairness mechanisms that approximate ex-post guarantees without excessively compromising economic performance constitutes an important avenue for future work.

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